Whither (Wither?) the Ultrasound Specialist?

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It was my great privilege earlier this year to be invited to give the Second Annual George R. Leopold, MD, Honorary Lecture at the Los Angeles Radiological Society's Spring Diagnostic Ultrasound Conference. This lectureship was initiated last year on his retirement from active practice as a clinician, researcher, teacher in ultrasound, and chairman of the Department of Radiology at the University of California (San Diego, CA).

As I contemplated the topic I would choose, I reflected on Dr Leopold's remarkably productive career. In addition to his own research advancing our field, he was also the teacher to many of our best-known current ultrasound teachers and researchers. He has truly fostered the growth and success of diagnostic sonography. In fact, his career spans and epitomizes the history of the clinical specialty of ultrasound from its infancy to the present. I fear that Dr Leopold, and the era he represented, may be irreplaceable. We, the generation of sonographic specialists he helped train, are in the latter years of our careers, with very few younger physicians coming along to replace us.

In this editorial, I will discuss the aspects of ultrasound that make it unique among diagnostic imaging modalities and why I believe this has led to a looming crisis for this specialty as it is performed in North America. My personal area of interest and expertise in ultrasound is in obstetrics, and I will limit my comments and concerns to this aspect of our field, although I suspect similar issues may exist in other sonographic subspecialties.

I want to emphasize that the problems I see facing us have absolutely nothing to do with "turf battles" between obstetricians and radiologists for the interpretation and control of obstetric sonography. It is very clear that the outstanding diagnosticians, researchers, and teachers in this subspecialty have come essentially equally from both these medical fields. Rather, it is the unique nature of ultrasound among the diagnostic imaging modalities that creates the problem.

The one major difference between ultrasound and other imaging disciplines was succinctly explained by Cindy Rapp, RDMS, an accomplished sonographer and excellent lecturer: "For radiology studies, the exam is easy, and the reading is difficult. For ultrasound, the exam is difficult, and the reading is easy."

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In most radiologic examinations, the image acquisition is done by prescribed protocols, straightforward and "easy." The analysis of the obtained images and their interpretation is the challenge. In sonography, the image acquisition is complex and operator dependent, and it becomes the diagnostic process. The subsequent reading confirms the completeness of the examination and validates the diagnosis but generally cannot go beyond the provided images. Thus, in large measure, the one who holds the transducer makes the diagnosis.

Another consequence of the real-time nature of sonographic diagnosis is that the more medical knowledge the sonographer or sonologist has, including pathophysiology as well as anatomy, the more intelligently that person can scan. He or she can seek out or exclude findings to support or rule out differential diagnostic possibilities. Merrill Sossman, MD, a former chief of radiology at the Peter Bent Brigham Hospital, stated this well: "You see what you look for and you look for what you know."

The physician will, in general, have more extensive and intensive education than the sonographer has in pathophysiology and the interactions between diseases and organ systems. Given the pertinent observations, the physician may well be able to make a more precise diagnosis than the sonographer.

The Crux of Our Developing Ultrasound Dilemma

The ideal sonographic diagnostician should have extensive medical knowledge and expert facility in performing the actual sonographic examination as well as interpreting it. That combination of training and skills is becoming progressively more rare among North American physicians, including radiologists and obstetricians. Both have less and less training time in ultrasound, and both have more and more conflicting duties, with little time for hands-on scanning.

Sonographers are our most adept scanners, but they have less depth of medical knowledge than physicians. They are also frequently being pushed to increase their throughput, completing ever more scans within their shifts.

In a real sense, the path to becoming an adept sonologist or sonographer is through the apprenticeship process. It requires hands-on scanning time and interpretation time but with both these tasks under the supervision of a mentor. Textbooks and lectures can provide a knowledge base, but they cannot replace this one-on-one mentoring process, and the greater the experience and skill of the teacher, the greater is the likelihood that the student will excel.

Over the past 10 to 20 years, practice patterns and pressures have been gradually eroding the effectiveness of this educational process. I believe we are today in North America in very real danger of losing our collective acumen in diagnostic ultrasound.

The Issues

Physician training in sonography during residency is quite limited. Radiology residents often have only 1 to 2 months of formal time in sonography, divided between general cases and those in obstetrics and gynecology. They frequently get only limited practice in the actual performance of the scans and tend to participate primarily in the interpretation of the studies, reading them followed by a review with a staff radiologist. They usually attend some case conferences. At a number of hospitals, obstetric sonography is done by the obstetrics service, and the radiology resident may have little or no access to these cases.

Residents in obstetrics and gynecology tend to learn sonography by performing "informal scans" on patients in the labor ward or in the clinic. Preserved images may be few or none; a staff physician may not get to check the scan; and a report may not be generated. That removes an important learning experience of committing one's interpretation to paper or to dictation and then having it reviewed and modified by an overreading supervisory physician (as usually occurs in radiology residencies). There may be limited exposure to technical training by sonographers, although perinatology fellows may have periods of assigned time doing scheduled scans on a sonographic service where sonographers and a supervising staff physician are present.

These patterns for training residents in radiology or obstetrics and gynecology have been prevalent for a decade or more. As a result, the attending staff mentors for these residents and fellows came up through the same systems of inadequate time, training, and caseload. As one cohort of future mentors with a pared-down intensity of exposure to a sonography caseload fails to learn or appreciate some of the subtleties and esoteric peculiarities of sonographic diagnosis, these details are lost to their future students. As these students go on to be mentors, the knowledge base they pass along may erode further.

Once the physicians have gone into practice, additional pressures tend to pull them away from active involvement in sonographic scanning. The real-time acquisition of images is seen as time-consuming, and it is often left to the sonographer alone. Competing duties are seen as more time-efficient and financially rewarding. The radiologist may favor time spent interpreting computed tomographic and magnetic resonance imaging scans and doing procedures such as biopsies. The obstetrician is frequently occupied with deliveries, procedures, and consultations.

Our sonographers tend to have the most methodical training in the performance of the examinations, and they see an adequate number of patients during their training. They are our most disciplined imagers, but it will be difficult for them to "bootstrap" themselves to better understanding of diagnosis and pathophysiology if their physician supervisors have limited training and experience in the nuances of sonographic interpretation.

There is objective evidence that expertise and experience make a difference in diagnostic accuracy, even within well-defined protocols. A 1994 report of the Routine Antenatal Diagnostic Imaging With Ultrasound Study trials compared the detection rate of fetal anomalies between tertiary (specialist) and nontertiary (generalist) diagnostic sonography practices. All sites used a standardized imaging protocol with examinations audited for completeness. The anomaly detection rate among tertiary sites was 39% versus only 13% at the nontertiary sites (detection likelihood ratio, 2.7).[1]

The problem with access to expert clinicians in sonographic diagnosis is compounded by the "graying of our mentors." Most of the speakers at North American review courses and symposia on ultrasound have been lecturing for 1 to 2 decades or more and had been practicing and researching before that to achieve their level of experience and knowledge. Some have retired, and many more of them will retire in the next 5 to 10 years. This would be fine and natural if new and rising "stars" were in the wings to replace them, but, unfortunately, there are remarkably few talented younger physicians choosing ultrasound as their primary specialty.

There has been, to my knowledge, only 1 dedicated ultrasound fellowship slot in the United States in the past decade. Radiologists with an interest in ultrasound tend to take fellowships either in cross-sectional imaging or women's imaging, and obstetricians gain advanced sonographic skills in maternal-fetal medicine fellowships, but all these incorporate ultrasound as a segment of their training only.

This long-term trend here has had a substantial negative impact on research and publication in diagnostic ultrasound. The clear fact is that North America once was but is no longer preeminent in the research advancement of our field. Current leading edge ultrasound research is, by and large, being conducted outside the United States and Canada. Over the past several years, at least half the articles published in the *Journal of Ultrasound in Medicine* and more than 90% of those in the journal *Ultrasound in Obstetrics and Gynecology* have come from countries elsewhere throughout the world.

It is my prediction that ultrasound in North America will become progressively less effective, both clinically and academically, unless we initiate a dramatic change in how sonography is practiced here.

A Paradigm Shift

I do not know whether the trends I have described can be reversed, but I offer 5 proposed directions that may help strengthen our ultrasound programs.

Mandatory Certification of Ultrasound Practices

Certification programs for various aspects of sonography are offered through 3 organizations: the American Institute of Ultrasound in Medicine for obstetric sonography, the American College of Radiology for obstetrics, gynecology, general, and the several vascular specialties, and the Intersocietal Commission for the Accreditation of Vascular Laboratories, also for the various types of vascular examinations.

These programs certify the sonography laboratories or practices rather than the individual physicians and sonographers performing and interpreting the examinations, but they do have training and continuing educational requirements for these individuals. These certifications are voluntary (with the exception that vascular certification is currently required for laboratories to be licensed in some states). There is recently published evidence that practices that seek and achieve accreditation improve their case study and guideline compliance scores when reevaluated 3 years after their initial applications.[2]

I believe that voluntary certification is a worthwhile start but that it is not adequate. Consider the Mammography Quality Standards Act enacted by the federal government. This act has substantially helped standardize and raise the quality of mammography, both by increasing the ongoing education of those reading these studies as a dedicated responsibility of their practice and also by forcing others who were unwilling to commit the time for study and quality control of mammography to drop that activity.

If ultrasound certification were made mandatory and were modeled on the Mammography Quality Standards Act, I believe it would similarly raise the bar. It would shift the caseload to practices that have made the effort to meet the requirements. It would also stop other practices that cannot or choose not to make the necessary educational and quality modifications from performing marginal or inadequate sonography.

Certificate of Added Qualifications in Ultrasound

Certificates of added qualifications (CAQs) have been developed by the American Board of Radiology (ABR) in 3 areas: pediatric radiology, neuroradiology, and vascular and interventional radiology. Radiologists, even without a fellowship, can demonstrate a level of advanced competence in these subspecialties by passing an oral examination, leading to a certification valid for 10 years, before renewal by a subsequent examination. Diagnostic sonography is a specialized area of imaging for which there is no current recognition for a physician. Were a CAQ offered, it might encourage a physician to prepare for and pass the examination to demonstrate that area of competency.

The current ABR CAQs are for an entire area of specialization. Many physicians with expertise in diagnostic ultrasound do not practice in all 3 segments of general, obstetric, and vascular sonography. A CAQ in sonography would almost certainly need to test a general familiarity with the global field of sonography and then evaluate a specific area of advanced abilities, such as obstetric sonography. The ABR CAQs have been offered only to radiologists. Perhaps the American College of Obstetricians and Gynecologists could develop a parallel and comparable certification process for sonography in obstetrics and gynecology. If this were a voluntary advanced certification, would physicians undertake the process, or would it be necessary to fold the CAQ in as an eventual requirement for continued accreditation of sonographic practices?

Centers of Comprehensive Sonographic Excellence

Throughout most of the world outside North America, sonography is practiced considerably differently from here. In most other countries, physicians are, by and large, the ones performing as well as interpreting the scans. For many or most of them, sonography is their primary medical professional activity. In a number of countries with socialized medicine, the government controls access to the practice of medicine. It may sanction only a single specialized provider center for each population area for a given medical activity such as diagnostic sonography or fetal medicine including sonographic services.

One prominent example of this approach is the Harris Birthright Center in London. This is a true center of excellence of international repute, combining clinical care, training, and research. The center is highly productive in all these areas. For example, under the guiding hand of its director, Kypros Nicolaides, MD, it has led the worldwide research in first-trimester nuchal translucency as an important risk predictor for aneuploidy and other fetal conditions. The center has gone further to create a program and certification process available throughout the world for performing this exacting measurement examination.

The United States does not have a socialized medical system to direct the organization of health care facilities, but the development here of one or two centers of excellence, at least for fetal medicine or women's imaging (and perhaps in other sonography-related areas also) would potentially create a nidus for the resurgence of clinical research, fellowship, and sonographer training in this country. If this concept is to have a chance to succeed, I believe it would require the cooperative efforts of the academic medical community, the national societies, including the American College of Radiology, American College of Obstetricians and Gynecologists, American

Institute of Ultrasound in Medicine, Society of Diagnostic Medical Sonography, and American Registry of Diagnostic Medical Sonographers, and the sonography equipment manufacturing industry, as well as financial support from government. Ideally, an equivalent center could be created in Canada.

Sonographic Practitioners

Sonographers are essential to the performance of diagnostic ultrasound examinations here, and that will remain so in the future. They receive the most methodical training in the technical skills necessary for the performance of high-quality diagnostic examinations. Unlike most physicians, they have no conflict of interest with other types of imaging studies or other clinical responsibilities.

I believe it is time to recognize the added benefit they could provide as physician extenders. Therefore, I support the creation of the new category of sonographic practitioner, analogous to the nurse practitioner or nurse anesthetist, who would be able to function independently within defined areas of responsibility, including rendering diagnoses and generating reports.

The Society of Diagnostic Medical Sonography has supported the formation of an Ultrasound Practitioner Commission, which has published a detailed proposal for such an advanced sonographic care provider.[3,4] The program would have entrance requirements equivalent to those of a registered diagnostic medical sonographer in any of the current disciplines. The accepted candidates would then matriculate in an educational program with in depth standardized clinical and didactic preparation. This would consist of a core curriculum covering some primary skills in all the sonographic disciplines as well as a more in-depth selected area of concentration: general, women's, vascular, or cardiac. The program would lead to a master's degree with certification through an advanced level national board examination. Ongoing continuing medical education appropriate to the selected area of clinical practice would be mandatory. (This model might also serve as a pathway for developing the physician CAQ in any one area of ultrasound.)

Technical Advances

Research and development in the ultrasound manufacturing industry will, no doubt, continue, and there are likely to be important improvements in 3- and 4-dimensional volume acquisitions. Eventually, a volume acquisition of a region or the entirety of a fetus could be formatted in multiple planes and presented much as magnetic resonance imaging is currently done. In the future, sophisticated computer programs may be able to select appropriate planes of reformatted images. There is already some prototype work that can automatically extract standard 2-dimensional image planes from a 4-dimensional acquisition of a single cardiac cycle.

When these technologies become clinically practical, they may allow physicians a greater degree of ability to interpret a scan from images alone, but this is certainly still unproven, and, for the near future, at least, I believe emphasis on diagnosis through real-time imaging will still predominate.

The Future: Whither or Wither?

I believe that the pattern of practice of sonography, as it has developed over the past 2 decades, leaves the field now at a precarious juncture. If we continue along the current path, the quality and clinical value of sonography will very likely continue to erode, and patient care and welfare will suffer. If, instead, we recognize the crisis and organize to pursue new solutions—those proposed here or others that arise through national dialogue among all our stakeholders—then we may see a new flourishing of diagnostic ultrasound.

References

1. Crane JP, LeFevre ML, Winborn RC, et al. A randomized trial of prenatal ultrasonographic screening: impact on the detection, management and outcome of anomalous fetuses. The RADIUS Study Group. Am J Obstet Gynecol 1994; 171:392–399.
2. Abuhamad AZ, Benacerraf BR, Woletz P, Burke BL. The accreditation of ultrasound practices: impact on compliance with minimum performance guidelines. J Ultrasound Med 2004; 23:1023–1029.
3. Hall R, Coffin C, Cyr D, et al. The ultrasound practitioner—a proposal: response to the SDMS for the development of a middle care provider in ultrasound imaging. J Diagn Med Sonography 1999; 15:140–156.
4. Hall R, Bierig M, Coffin C, et al. Ultrasound practitioner master's degree curriculum and questionnaire response by the SDMS membership. J Diagn Med Sonography 2001; 17:154–161.